Solar Charging Station



Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Solar charging station

What it is used for

A solar charging station converts solar energy into electrical power for charging compatible batteries or devices. Operate it within its rated voltage and current limits and follow the installed system's electrical protection and charging instructions.

Main parts

- Solar PV panels and mounting structure
- Charge controller or power conversion unit
- Battery storage system, if fitted
- Charging outlets, connectors, or charging ports
- Protection devices, isolators, and circuit breakers
- Monitoring display and earthing system

Before you start

1. Confirm the station's output ratings and compatibility with the intended battery or device.
2. Inspect panels, mounting, cables, connectors, enclosures, and charging ports for damage.
3. Check that earthing, protective devices, and isolators are installed and accessible.
4. Ensure the area around the station is clear, dry, and adequately ventilated.
5. Check battery condition and follow the battery manufacturer's charging requirements.
6. Confirm that all connections and protection settings match the approved installation design.

How to use

1. Keep the charging output switched off while inspecting or connecting equipment.
2. Position the vehicle or device at the designated charging point.
3. Verify the battery or device voltage and charging requirements against the station rating.
4. Connect the approved charging cable securely to the correct port.
5. Switch on the required isolator or controller according to the installation procedure.
6. Start charging using the approved control or user interface.
7. Check the display for charging status, current, voltage, and warning indicators.
8. Monitor the system for unusual heat, smell, noise, or error messages.
9. Stop charging if a fault, damaged connector, or abnormal condition occurs.
10. Complete charging according to the battery or device instructions.
11. Stop the session using the approved control and disconnect the cable safely.
12. Switch off the charging output if required and return cables to their holders.
13. Record faults or abnormal readings and report them to the responsible technician.

⚠ Safety

- Only qualified personnel should install, repair, or service the electrical system.
- Solar panels may remain live in daylight; never touch exposed conductors or connectors.
- Do not charge damaged, swollen, leaking, or incompatible batteries.
- Keep connectors dry and do not plug or unplug damaged connectors under load.
- Never bypass fuses, circuit breakers, earthing, or other protective devices.
- Follow battery-specific fire procedures and keep the required emergency access clear.

Care & storage

- Clean panels using the approved method and safe isolation procedure.
- Inspect cables, connectors, mounting structures, and protective devices regularly.
- Keep charging ports clean, dry, and protected from impact.
- Check controller settings and battery condition according to the maintenance schedule.
- Arrange qualified service for faults, corrosion, overheating, or repeated charging errors.

Common mistakes to avoid

- Connecting a battery or device with incompatible charging requirements.
- Ignoring damaged cables, connectors, or warning indicators.
- Touching PV wiring without qualified electrical precautions.
- Bypassing protective devices or leaving faulty charging equipment in use.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Prepared by - Subodh Kumar